Obstructions and a Corrected Spectral Criterion for GRH

GRH Research Project
Working manuscript; authorship to be confirmed

November 2, 2026

Research draft. GRH is not proved here.

Abstract

We audit proposed surface-order and Hilbert–Pólya arguments for GRH. We retain a conditional spectral-approximation criterion and prove three obstructions: integral hyperbolic matrices have no integer prime-power norms, integral cyclic induction fails already for the Klein four group, and absolute prime tails diverge throughout the proposed critical-strip range. We also exhibit determinant information lost by symmetrization and a direct failure of the supplied diagonal diagnostic. We supply convergent arithmetic replacements using Dirichlet partial sums and pole-subtracted zeta sums, with explicit errors. These results identify what a complete arithmetic spectral construction must still establish. They do not establish GRH. A Lean 4 companion verifies exact arithmetic obstructions, complex Euler-factor nonvanishing, and a local boundary-convergence lemma, spectral matrix criteria and Jensen metric components. The full arithmetic spectral bridge remains unresolved.

1 Scope and conventions

For a primitive Dirichlet character χ of conductor q, write

$$L(s,\chi)=\sum_{n\ge 1}\frac{\chi(n)}{n^s},\qquad \Lambda(s,\chi)=\left(\frac{q}{\pi}\right)^{(s+a)/2}\Gamma\left(\frac{s+a}{2}\right)L(s,\chi),\quad a=\frac{1-\chi(-1)}{2}.$$

The series and Euler product initially apply for $\operatorname{Re} s>1$. Our target is Dirichlet GRH: every zero in $0<\operatorname{Re} s<1$ has real part $1/2$. For $q=1$, use the entire function $\xi(s)=\frac{1}{2}s(s-1)\Lambda(s,1)$ when an entire target is required. Finite-order Hecke characters and general Artin representations require separate constructions; a theorem for one family is not silently a theorem for another.

The supplied surface-order manuscripts [1, 2] propose arithmetic transfer operators and self-adjoint approximants. We isolate valid closure logic and prove obstructions to particular claimed constructions. These obstructions refute intermediate statements, not GRH. The separate `math` repository [3] includes claimed zero-free regions strictly weaker than GRH. Such a region cannot replace the missing critical-line argument.

2 A corrected conditional closure theorem

Theorem 2.1 (Spectral approximation criterion)**.** *Let $U\subset\mathbb{C}$ be connected and open, let f be holomorphic on U and not identically zero, and let*

$$F_N(s)=e^{a_N+b_Ns}\det\bigl(sI-(\tfrac{1}{2}I+iH_N)\bigr),\qquad H_N=H_N^*\in M_{d_N}(\mathbb{C}).$$

If $F_N\to f$ locally uniformly on U, then every zero of f in U has real part $1/2$.

Proof. The eigenvalues $\lambda_{j,N}$ of H_N are real. Thus the zeros of F_N are $1/2 + i\lambda_{j,N}$, with algebraic multiplicities; the exponential factor never vanishes. If $f(z) = 0$ off that line, choose a closed disk $D \subset U$ about z disjoint from the line, with f nonzero on its boundary. Such a disk exists because the zeros of a nonzero holomorphic function on a connected domain are isolated. Let $m = \min_{\partial D} |f| > 0$. Local uniform convergence gives $|F_N - f| < m$ on ∂D for sufficiently large N. Rouché's theorem forces F_N and f to have the same positive number of zeros inside D, contradicting the zero location of F_N. □

Remark 2.2. This is a conditional criterion, not a construction. Self-adjointness alone establishes only the zero location of the approximants. The difficult arithmetic statement is convergence to the particular completed L-function. Pointwise convergence, weak operator convergence, and agreement of finitely many zeros do not supply the displayed hypothesis.

Proposition 2.3 (A usable continuation interface)**.** *Suppose U is a connected domain, the entire functions F_N are locally uniformly bounded on U, and they converge pointwise on a nonempty open subset $V \subset U$ to a holomorphic function f defined on U. Then $F_N \to f$ locally uniformly on U.*

Proof. Montel's theorem makes the family normal. Every subsequence has a further subsequence converging locally uniformly to a holomorphic function g. On V, pointwise convergence gives $g = f$, so the identity theorem gives $g = f$ on U. If the original sequence failed to converge locally uniformly, a compact set, positive error, and violating subsequence would contradict this subsequence property. □

A proposed analytic continuation proof must actually provide the uniform bounds and a genuine overlap V. The disjoint domains $\operatorname{Re} s > 1$ and $0 < \operatorname{Re} s < 1$ are not an overlap.

3 Arithmetic obstructions to the claimed completion

Theorem 3.1 (No integral prime-power geodesic norms)**.** *Let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ be hyperbolic. Its expanding eigenvalue in absolute value is $u > 1$, and its hyperbolic norm is $N(\gamma) = u^2$. Then $N(\gamma)$ is not an integer greater than one. In particular no closed geodesic arising from such a matrix has length $k \log p$ for a prime p and integer $k \geq 1$.*

Proof. Put $t = |\operatorname{tr} \gamma| \in \mathbb{Z}$, so $t = u + u^{-1}$. If $u^2 = n \in \mathbb{Z}$, $n > 1$, squaring gives

$$t^2 = n + 2 + \frac{1}{n}.$$

The left-hand side and $n + 2$ are integers, so $1/n$ is an integer, contradicting $0 < 1/n < 1$. Since length is $\log N(\gamma)$, length $k \log p$ would imply $N(\gamma) = p^k$, already excluded. □

The statement applies to every subgroup of $\mathrm{SL}_2(\mathbb{Z})$, hence to congruence surfaces. The prime geodesic theorem is an asymptotic counting analogy; it cannot provide the termwise identification claimed in the completion section of [1]. Hecke correspondences of determinant p must not be identified with hyperbolic deck transformations of determinant one.

Theorem 3.2 (Failure of integral cyclic induction)**.** *The trivial character of $G = C_2 \times C_2$ is not an integer linear combination of characters induced from one-dimensional characters of cyclic subgroups of G.*

Proof. Every cyclic subgroup has order one or two. The degree of any such induced character is its index, namely four or two. Evaluation at the identity sends every integer linear combination to an even integer, whereas the trivial character has degree one. □

This distinguishes rational Artin induction from integral Brauer induction. It invalidates the stated cyclic integral reduction, while leaving appropriate forms of these classical induction theorems available for a revised argument.

Theorem 3.3 (Prime-tail obstruction)**.** *For every real $\sigma \le 1$, the positive series $\sum_p p^{-\sigma}$ diverges. Thus $\sum_{p>N} p^{-\sigma}$ is not a vanishing absolutely convergent tail in the critical strip.*

Proof. It suffices to show $\sum_p 1/p = \infty$, since $p^{-\sigma} \ge p^{-1}$. Suppose the reciprocal-prime series converged. For $0 \le x \le 1/2$, $-\log(1-x) \le 2x$, for example by integrating $(1-t)^{-1} \le 2$ from zero to x. Hence the finite Euler products $\prod_{p \le X}(1-1/p)^{-1}$ would be bounded uniformly in X. Expanding each as a finite product of absolutely convergent geometric series gives the sum of $1/n$ over all positive integers with prime factors at most X. In particular it dominates $\sum_{n \le \lfloor X \rfloor} 1/n$, which is unbounded. Contradiction. □

Any repair below $\operatorname{Re} s = 1$ must introduce proved cancellation or a regularization, with quantitative control; absolute prime-tail bounds cannot suffice.

4 Determinants and lost arithmetic information

For $T = \operatorname{diag}(\chi(p)p^{-s})_{p \le N}$ one has exactly

$$\det(I - T) = \prod_{p \le N} (1 - \chi(p)p^{-s}) = L_N(s, \chi)^{-1},$$

not L_N. On an absolute-convergence region,

$$\log\det(I - T) = -\sum_{k \ge 1} \frac{\operatorname{tr}(T^k)}{k}.$$

The reciprocal orientation matters before any analytic continuation. Moreover Hermitian-part extraction does not preserve this determinant:

$$T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \frac{T + T^*}{2} = 0, \quad \det(I - T) = 2 \ne 1 = \det(I).$$

An identity for the transfer determinant therefore does not automatically identify the centered characteristic polynomial of a symmetrized operator.

Proposition 4.1 (A direct obstruction to the supplied diagonal diagnostic)**.** *For the $q = 1$ model evaluated at $s = 1/2$, the algorithm that sets $\lambda_p = -\arg(p^{-1/2})$ gives $H_N = 0$ and $F_N(s) = (s - 1/2)^{d_N}$. These functions cannot converge locally uniformly to $\Lambda(s, 1)$ or $\xi(s)$ on a neighborhood of $1/2$.*

Proof. Every $p^{-1/2}$ is positive, so its argument is zero and $F_N(1/2) = 0$. On the other hand the alternating Dirichlet series $\eta(1/2) = \sum_{n \ge 1}(-1)^{n-1}n^{-1/2}$ is positive: grouping into adjacent pairs gives positive terms and its sum is at least $1 - 1/\sqrt{2} > 0$. The continuation identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ gives $\zeta(1/2) < 0$. The gamma factor at $1/4$ is positive and finite, so $\Lambda(1/2, 1) \ne 0$, and its pole-removal factor at $1/2$ is nonzero. Even pointwise convergence at that point is impossible. □

Proposition 4.2 (Holomorphic Hermitian rigidity)**.** *Let $U \subset \mathbb{C}$ be connected and open, with a fixed inner product on $\mathbb{C}^d$. If $H : U \to M_d(\mathbb{C})$ has holomorphic entries and $H(s)^* = H(s)$ for every complex $s \in U$, then H is constant.*

Proof. In fixed orthonormal coordinates, for each vector v, the holomorphic scalar $v^*H(s)v$ is real-valued, so it is constant by the open mapping theorem. Polarization recovers all matrix entries from these quadratic forms, hence every entry is constant. Equivalently, $H_{ij} + H_{ji}$ and $i(H_{ij} - H_{ji})$ are real-valued holomorphic functions whose constancy determines H_{ij}. □

Thus a nonconstant arithmetic transfer $T(s)$ cannot retain entrywise holomorphy after pointwise Hermitian-part extraction: a holomorphic $(T + T^*)/2$ would be constant, and the remaining holomorphic skew-Hermitian part would be constant as well. This excludes that particular shortcut, not fixed self-adjoint operators. The spectral criterion uses H_N independent of the spectral variable s. Requiring self-adjointness only on a real parameter line is also consistent with a nonconstant holomorphic extension, but needs its own zero-location theorem.

Proposition 4.3 (A positive generalized spectral pencil)**.** *Let $A = A^*$ and $B = B^* > 0$ be fixed $d \times d$ matrices. Every zero of $\det(A - zB)$ is real. Consequently every zero of $\det(A + i(s - 1/2)B)$ lies on $\operatorname{Re} s = 1/2$.*

Proof. Singularity supplies $v \neq 0$ with $Av = zBv$. Hence $z = (v^*Av)/(v^*Bv)$, with real numerator and strictly positive real denominator. Thus z is real. Substitute $z = -i(s - 1/2)$. □

This supplies a valid finite spectral mechanism, but not its arithmetic construction. To apply it one must construct A_N, B_N, certify $B_N > 0$, and prove the required completed-L determinant approximation independently. Hermitian coefficients alone are insufficient: for

$$A = \begin{pmatrix} 0 & \varepsilon \\ \varepsilon & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \det(A - zB) = -(z^2 + \varepsilon^2),$$

$0 < \varepsilon < 1/2$ gives transported off-line zeros $s = 1/2 \pm \varepsilon$ inside the critical strip.

5 A necessary change of approximation class

Theorem 5.1 (Zero-free Euler truncations cannot approximate a zero)**.** *Let U be a connected open subset of $\{\operatorname{Re} s > 0\}$, let g_N be holomorphic and nowhere zero on U, and let P_N be arbitrary finite sets of rational primes. Put*

$$D_N(s) = \prod_{p \in P_N} (1 - \chi(p)e^{-s \log p}).$$

Each of $g_N D_N$ and g_N / D_N is holomorphic and nowhere zero on U. If either sequence converges locally uniformly to a holomorphic f which is not identically zero, then f has no zero in U.

Proof. For $\operatorname{Re} s > 0$, $|\chi(p)e^{-s \log p}| \leq p^{-\operatorname{Re} s} < 1$, so every local factor is nonzero. Hence the finite product, its reciprocal, and their normalized versions have the asserted property. If f had a zero, choose a closed disk inside U with a positive number of zeros of f in its interior and none on its boundary. Local uniform convergence and Rouché's theorem would force the zero-free approximants to have that positive number of zeros, a contradiction. □

More quantitatively, on any such disk D, every zero-free holomorphic approximant G satisfies

$$\max_{\partial D} |G - f| \;\geq\; \min_{\partial D} |f| > 0.$$

Otherwise Rouché yields the same contradiction. Thus improved estimates cannot repair this approximation class near a nontrivial zero. The construction must change so that finite approximants can have zeros in the strip; a nonvanishing gamma or exponential normalization cannot do this.

6 A convergent arithmetic replacement

For a nonprincipal character modulo q, put $A(x) = \sum_{n \leq x} \chi(n)$ and $B = \sup_x |A(x)| \leq q$. Complete periods sum to zero, so B is finite. Partial summation gives, for $\sigma = \operatorname{Re} s > 0$,

$$L(s,\chi) - \sum_{n \leq N} \chi(n) n^{-s} = -A(N)N^{-s} + s \int_N^\infty A(x) x^{-s-1}\, dx,$$

and therefore

$$\left| L(s,\chi) - \sum_{n \leq N} \chi(n) n^{-s} \right| \leq B \left(1 + \frac{|s|}{\sigma}\right) N^{-\sigma}.$$

The endpoint tends to zero and the integral is absolutely convergent. The limit agrees with the Euler definition on $\operatorname{Re} s > 1$, hence with its analytic continuation. Multiplication by the correct gamma/conductor factor gives locally uniform approximation on $\operatorname{Re} s > 0$ for either parity.

For zeta, reproduce the pole before removing it. Set

$$Z_N(s) = \sum_{n \leq N} n^{-s} + \frac{N^{1-s}}{s-1}.$$

Euler summation gives, for $\sigma > 0$ and $s \neq 1$,

$$\zeta(s) - Z_N(s) = -s \int_N^\infty \{x\} x^{-s-1}\, dx, \qquad |\zeta(s) - Z_N(s)| \leq \frac{|s|}{\sigma} N^{-\sigma}.$$

The integral continues holomorphically to $\sigma > 0$. Consequently

$$\xi_N(s) = \frac{s}{2} \pi^{-s/2} \Gamma(s/2) \left((s-1) \sum_{n \leq N} n^{-s} + N^{1-s} \right)$$

is holomorphic on that half-plane, including $s = 1$, and converges locally uniformly to ξ; in fact $\xi_N(1) = 1/2$ for every N. These constructions use known arithmetic coefficients, not unknown zeros. They avoid Theorem 5.1 by allowing additive cancellation and unrestricted finite-level zeros. They establish convergence, but provide no self-adjoint realization or critical-line theorem for those zeros.

7 A certified test of the arithmetic pencil candidate

Let χ be the primitive nonprincipal character modulo 3. Its fifth Dirichlet partial sum is the entire function

$$f(s) = S_5(s,\chi) = 1 - 2^{-s} + 4^{-s} - 5^{-s}.$$

Theorem 7.1 (Certified zero of a finite partial sum)**.** *Put*

$$c = (337261 + 11027504i)/10^6, \qquad r = 10^{-4}.$$

The disk $|s - c| < r$ *contains exactly one zero of* f*, counted with multiplicity, and lies inside* $0 < \operatorname{Re} s < 1/2$*.*

Proof. The exact rational procedure detailed in the extended edition establishes

$$|f(c)| < 10^{-6}, \qquad |f'(c)| > 1, \qquad \sup_{|s-c|\le r} |f''(s)| \le 9.$$

It uses 60 terms of the logarithm's positive atanh series and terms 0 through 120 of the exponential Taylor series, with explicit remainders; no floating-point value is used in these inequalities. On the boundary, Taylor's formula gives

$$|f(s) - f'(c)(s-c)| < 10^{-6} + \tfrac{9}{2}r^2 < r < |f'(c)(s-c)|.$$

Rouché gives exactly one zero, which is therefore simple. The real part throughout the closed disk belongs to $[337161, 337361]/10^6 \subset (0, 1/2)$. □

The completed fifth partial sum has the same zero because its conductor/gamma factor is holomorphic and nonzero on this disk. It cannot equal a nowhere-zero holomorphic factor times the determinant of a fixed positive Hermitian pencil there. This rejects that exact finite-level identification. It does not reject convergence of the partial sums, modified approximants, later truncations, subsequences, or GRH itself: f is a finite partial sum, not the full L-function.

8 Jensen approximants and a coefficient-defined metric

This alternative currently concerns the Riemann target only. Define

$$\gamma_j = \frac{j!\,\xi^{(2j)}(1/2)}{(2j)!}, \qquad F(w) = \sum_{j\ge 0} \frac{\gamma_j}{j!} w^j, \qquad \Xi(t) = \xi(1/2 + it) = F(-t^2).$$

The classical positive Fourier kernel of Ξ gives $\gamma_j > 0$ [9]; evenness and entirety of ξ make F entire. Set, for $d \ge 1$,

$$J_d(x) = \sum_{j=0}^{d} \binom{d}{j} \gamma_j x^j, \qquad Q_d(w) = J_d(w/d), \qquad P_d(t) = Q_d(-t^2).$$

Proposition 8.1 (Explicit convergence without a zero-location premise)**.** *For* $R, T \ge 0$*,*

$$\sup_{|w|\le R} |Q_d(w) - F(w)| \le \frac{R^2 F''(R)}{2d}, \qquad \sup_{|t|\le T} |P_d(t) - \Xi(t)| \le \frac{T^4 F''(T^2)}{2d}.$$

Proof. The coefficient multiplier $w_{d,j} = j!\binom{d}{j}/d^j$ for $j \le d$, and 0 for $j > d$, satisfies $0 \le w_{d,j} \le 1$ and $1 - w_{d,j} \le j(j-1)/(2d)$. For $j \le d$ use $1 - \prod_{k<j}(1 - k/d) \le \sum_{k<j} k/d$; for $j > d$ use $j(j-1) \ge 2d$. Summing the absolutely convergent series with nonnegative coefficients proves the bounds. □

Let C_d be the multiplication-by-x companion of the monic J_d/γ_d, and $H_d = (m_{i+j})_{0\le i,j<d}$, where $m_k = \operatorname{tr}(C_d^k)$ and $m_0 = d$. Newton identities compute H_d from the coefficients, and $H_d C_d = C_d^T H_d$. The Hermite–Sylvester criterion gives $H_d \ge 0$ exactly when J_d is real-rooted [8]; then its positive coefficients force all roots negative. If $H_d > 0$, the extended edition constructs an explicit real symmetric K_d with

$$P_d(t) = \frac{(-1)^d \gamma_d}{d^d} \det(tI - K_d).$$

Strict positivity is needed for that construction; semidefinite metrics can be singular. The classical Jensen criterion makes $H_d \ge 0$ for every d equivalent to zeta RH [7]. This identifies the missing positivity; it does not prove it. Fixed-degree eventual hyperbolicity as the shift grows does not settle this unshifted growing-degree family.

Proposition 8.2 (Positive smooth kernels do not supply the metric)**.** *Positivity of an even smooth Fourier density, including Gaussian decay, does not imply positivity of its Jensen Newton-sum metric.*

Proof. Convolve $\frac{9}{10}\delta_0 + \frac{1}{20}(\delta_{-1} + \delta_1)$ with a centered Gaussian of variance $v > 0$. Its Fourier transform is $e^{-vt^2/2}(9/10 + \cos t/10)$ and its density is strictly positive, even and Schwartz. Writing its associated $F(w) = \sum \gamma_j w^j/j!$ gives $\gamma_0 = 1$, $\gamma_1 = (v + 1/10)/2$ and $\gamma_2 = (3v^2 + 3v/5 + 1/10)/12$. Thus $\gamma_1^2 - \gamma_0\gamma_2 = -7/1200 < 0$; the degree-two Newton-sum metric has negative determinant. These are model coefficients, not the arithmetic ξ coefficients. □

9 Consequences for a revised research article

Theorems 3.1, 3.2, and 3.3 require replacements for the associated proof steps, not additional exposition. A credible next theorem would construct operators independently of unknown zeros and establish their exact arithmetic determinant, including ramified, archimedean, and conductor terms. It must then prove locally uniform convergence or the local boundedness in Proposition 2.3. Only then does Theorem 2.1 prove the desired zero location.

Weil/Li positivity, adelic trace methods, and canonical-product approximations are potential alternative interfaces. They move the research obligation to a specific positivity, trace, or approximation theorem; they do not discharge it by analogy. Lagarias [4] and Connes [5] provide relevant primary-source starting points. No novelty or complete proof is claimed for these classical conditional principles.

References

[1] A. Schreiber, *Emergent Cyclic Time from Surface Orders and a Hilbert–Pólya Program for the Generalized Riemann Hypothesis*, repository source, commit `212ff730`, `https://github.com/amelie-iska/Surface-Orders-Hilbert-Polya`.

[2] A. Schreiber, *Surface-Order Determinants, Monomial Artin L-Functions, and the General Hilbert–Pólya Extension Problem*, repository PDF, commit `0c77122d`, `https://github.com/amelie-iska/GRH`.

[3] *Mathematical manuscripts and supporting proof artifacts*, supplied repository, `https://github.com/amelie-iska/math`. See project source manifest for the exact local pin and review scope.

[4] J. C. Lagarias, Li coefficients for automorphic L-functions, *Ann. Inst. Fourier* **57** (2007), 1689–1740, https://arxiv.org/abs/math/0404394v4.

[5] A. Connes, Trace formula in noncommutative geometry and the zeros of the Riemann zeta function, *Selecta Math.* **5** (1999), 29–106, https://arxiv.org/abs/math/9811068.

[6] The mathlib community, *Riemann zeta function documentation*, https://leanprover-community.github.io/mathlib4_docs/Mathlib/NumberTheory/LSeries/RiemannZeta.html.

[7] M. Griffin, K. Ono, L. Rolen and D. Zagier, Jensen polynomials for the Riemann zeta function and other sequences, *Proc. Natl. Acad. Sci. USA* **116** (2019), 11103–11110, https://doi.org/10.1073/pnas.1902572116.

[8] M. B. Nathanson, The Hermite–Sylvester criterion for real-rooted polynomials, version 2 (2019), https://arxiv.org/abs/1911.01745v2.

[9] D. Romik, Orthogonal polynomial expansions for the Riemann xi function, version 3 (2019), equations 1.8–1.11, https://arxiv.org/abs/1902.06330v3.